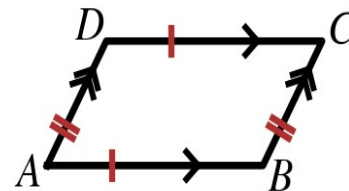


Speak Like A Mathematician:

quadrilateral:

parallelogram:

diagonals of a quadrilateral:



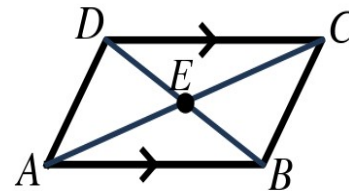
Properties of Parallelograms

If a quadrilateral is a parallelogram, then: opposite sides are _____.
opposite angles are _____. consecutive angles are _____. the
diagonals _____ each other.

Example 1 - Prove Opposite Sides Are Congruent

Given: ABCD is a parallelogram with diagonal $\overline{AC}$

Prove: $\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$

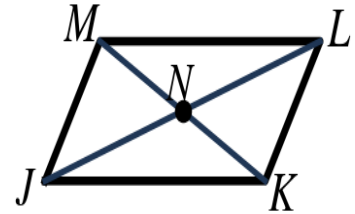


Statements	Reasons
1. ABCD is a parallelogram with diagonal $\overline{AC}$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.

Example 2 - Use the Properties

In $\square JKLM$ at the right, the diagonals meet at N . $KL = 58$ mm, $KN = 26$ mm, and $m\angle MJK = 51^\circ$.

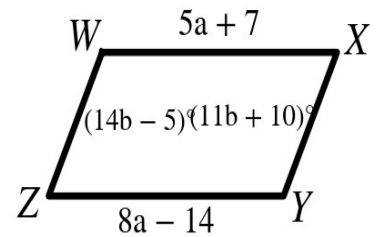
- Find MJ .
- Find $m\angle JML$.
- Find $m\angle KLM$.
- Find KM .



Example 3 - Solve for the Variables

$WXYZ$ at the right is a parallelogram.

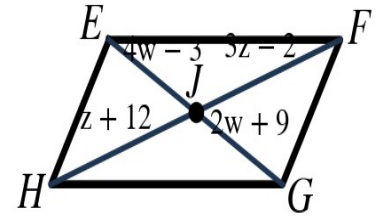
- Find a , then find ZY .
- Find b , then find $m\angle X$.
- Find $m\angle Z$.



Example 4 - Diagonals

In $\square EFGH$ at the right, the diagonals meet at J . $EJ = 4w - 3$, $JG = 2w + 9$, $FJ = 3z - 2$, and $JH = z + 12$.

- Find w .
- Find z .
- Find EG and FH .



Example 5 - Find the Fourth Vertex

Three vertices of $\square PQRS$ are $P(-4, -1)$, $Q(-2, 5)$, and $S(4, 1)$. Find vertex R .

- Which segment must be parallel and congruent to $\overline{PQ}$?
- Use the 'same rise, same run' idea to find R .
- Check with the diagonals.

